

A Ruppeiner curvature dichotomy for generalized prime gases: asymptotic flatness at limiting-temperature transitions

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June 14, 2026

Abstract

We study the thermodynamic (Ruppeiner) geometry of grand-canonical *generalized prime gases*, whose partition functions are Dirichlet series $\log \Xi(\beta, z) = \sum_{k \geq 1} (z^k/k) \mathcal{P}(k\beta)$. We prove that the scalar curvature at a phase transition is governed by *where the singularity lives*: a *temperature-driven* (Hagedorn / limiting-temperature) transition forces the curvature to vanish asymptotically, $R \rightarrow 0$, while a *fugacity-driven* (condensation) transition makes it diverge, $|R| \rightarrow \infty$. The discriminating mechanism is a single algebraic condition on the singular sector of $\log \Xi$ — it is curvature-flat precisely when that sector factorizes as $e^y \varphi(x)$ in the natural coordinates, i.e. when fugacity and temperature do not couple. The result is established by an explicit cancellation lemma together with an analyticity argument, and is bracketed by four exactly-computable instances: the classical ideal gas ($R = 0$), the Riemann primon gas and an all-integer Bose gas (both Hagedorn, $R \rightarrow 0$), and the ideal Bose gas at condensation ($|R| \rightarrow \infty$). Two of the four contain no primes; the statement is one of statistical mechanics.

1 Introduction

The Ruppeiner scalar curvature R of the thermodynamic state space offers a geometric measure of underlying statistical fluctuations, classically diverging at critical points where the correlation volume becomes macroscopic [5]. For one-dimensional manifolds such curvature claims are vacuous; in a two-parameter (grand-canonical) ensemble, however, R is a non-trivial invariant [5, 2].

In this paper we investigate the thermodynamic geometry of grand-canonical generalized prime gases (Beurling gases) near their continuous phase transitions. For the canonical primon gas, the Hagedorn temperature ($\beta = 1$) induces a well-known divergence in the partition function [3, 4, 6, 1]. We extend this to a two-parameter grand-canonical manifold evaluated along the locking ray $z = 1$.

We establish a rigorous curvature dichotomy: for generalized prime gases characterized by a definite leading singularity, the scalar curvature R strictly vanishes (asymptotically flat) at temperature-driven (Hagedorn-like) singularities, but diverges ($|R| \rightarrow \infty$) at fugacity-driven (Bose–Einstein condensation) transitions. We prove that the flat sector is unconditional within this class, driven by an exact decoupling in the Hessian determinant.

2 Setup

Definition 1 (generalized prime gas). A grand-canonical gas with

$$\log \Xi(\beta, z) = \sum_{k \geq 1} \frac{z^k}{k} \mathcal{P}(k\beta), \tag{1}$$

where $\mathcal{P}$ is the spectral (prime) zeta of the mode spectrum, β the inverse temperature and z the fugacity.

Use natural coordinates $(x, y) = (-\beta, \log z)$. The family is exponential with sufficient statistics (E, N) , so the Ruppeiner/Fisher metric is the Hessian of the potential $\psi := \log \Xi$,

$$g = \text{Hess}_{(x,y)} \psi, \quad \partial_x^a \partial_y^b \psi = (-1)^a \sum_{k \geq 1} k^{a+b-1} z^k \mathcal{P}^{(a)}(k\beta). \quad (2)$$

For a two-dimensional Hessian metric the scalar curvature is

$$R = -\frac{N}{2(\det g)^2}, \quad N = \det \begin{pmatrix} \psi_{xx} & \psi_{xy} & \psi_{yy} \\ \psi_{xx} & \psi_{xy} & \psi_{yy} \\ \psi_{xx} & \psi_{xy} & \psi_{yy} \end{pmatrix}. \quad (3)$$

Remark 1 (convention). We use the sphere-positive scalar convention, in which the normal family $\psi = -t_1^2/(4t_2) - \frac{1}{2} \log(-2t_2)$ returns $R = -1$ at every point; (3) reproduces this and the flat-product control $R = 0$. Only $|R|$ (vanishing vs. divergence) is convention-independent; reported signs are fixed by this choice.

Hypotheses.

- (H1) *Temperature-driven singularity.* $\mathcal{P}(s)$ is real-analytic on $(1, \infty)$ with an isolated singularity at its abscissa $s = 1$ and no other singularity on the real axis $s \geq 1$. (The transition is a singularity in β at the limiting temperature $\beta = 1$.)
- (H2) *Finite background.* $\sum_{k \geq 2} k(k-1)\mathcal{P}^{(j)}(k) < \infty$ for $j = 0, 1, 2$ (automatic when $\mathcal{P}(k) \rightarrow 0$ geometrically).
- (C1) *Definite leading order.* The abscissa singularity is of pole type $A\varepsilon^{-\alpha}$ ($\alpha > 0$) or logarithmic. (Holds for every natural Beurling/prime spectral zeta.)

3 The product-form criterion

Lemma 1 (product form annihilates the curvature numerator). *If $\psi_{\text{sing}} = e^y \varphi(x)$ for a smooth φ , then the third row of N in (3), $(\psi_{xxy}, \psi_{xyy}, \psi_{yyy})$, equals the first row $(\psi_{xx}, \psi_{xy}, \psi_{yy})$ identically and at all orders; hence the contribution of ψ_{sing} to N is $\equiv 0$.*

Proof. $\partial_y(e^y \varphi^{(j)}(x)) = e^y \varphi^{(j)}(x)$, so each entry of row three equals the corresponding entry of row one. A determinant with two equal rows vanishes; the identity is exact in ε , hence order by order. $\square$

Lemma 2 (only the single-particle term is singular). *Under (H1), every $k \geq 2$ term of (1) is real-analytic at $(\beta, z) = (1, 1)$, and the unique singular term is $\psi_1 = z\mathcal{P}(\beta) = e^y \varphi(x)$, $\varphi(x) := \mathcal{P}(-x)$.*

Proof. For $k \geq 2$ and β near 1, $\mathcal{P}(k\beta)$ is evaluated at $k\beta \approx k \geq 2$, inside the analytic region by (H1); the term is analytic. Only $k = 1$ probes the abscissa, and it is linear in z . The product form is forced by the singularity living in temperature and being sampled at the integer multiples $k\beta$. $\square$

4 The dichotomy theorem

Theorem 1 (temperature-driven $\Rightarrow$ asymptotically flat). *Under (H1)–(H2) and (C1), as $(\beta, z) \rightarrow (1, 1)$ along $z = 1$, $R \rightarrow 0$. Explicitly, with $L = \log(1/\varepsilon)$, $\varepsilon = \beta - 1$, background amplitude $\Delta_3 := \sum_{k \geq 2} k(k-1)z^k \mathcal{P}(k)$, and leading singularity $\mathcal{P}(1+\varepsilon) \sim A\varepsilon^{-\alpha}$,*

$$\alpha > 0 : R \sim \frac{\Delta_3(\alpha+1)}{2A^2} \varepsilon^{2\alpha}, \quad \alpha \rightarrow 0 \text{ (logarithmic)} : R \sim \frac{\Delta_3}{2(L+\kappa)^2}.$$

Proof. By Lemma 2 the singular sector is $\psi_1 = e^y \varphi(x)$; by Lemma 1 it contributes 0 to N , so N is sourced entirely by the analytic $k \geq 2$ background. The full cofactor reduction and the resulting rates are given in Appendix A; the flatness conclusion is unconditional within the class (Remark 4). $\square$

Proposition 1 (fugacity-driven $\Rightarrow$ divergent). *If the transition is at $z \rightarrow z_c$ with the singularity carried by the full fugacity series, $\log \Xi_{\text{sing}} = \varphi(\beta) \text{Li}_a(z)$, then the singular sector $\varphi(\beta) \text{Li}_a(e^y)$ is not of the form $e^y \varphi(x)$; Lemma 1 does not apply, $\text{Row}_3 \neq \text{Row}_1$, and $|R| \rightarrow \infty$.*

Remark 2. Theorem 1 and Proposition 1 give the discriminant in one line:

$$R \rightarrow 0 \iff \text{the singular sector of } \log \Xi \text{ factorizes as } e^y \varphi(x).$$

5 Instances

All four instances run through the same engine (3); the classical ideal gas validates it ($R = 0$ is forced and obtained).

gas	singular structure	type	curvature (rate / amplitude)
classical ideal gas	$u^{-3/2} e^y (\varphi \cdot e^y)$	product form	$R = 0$ (to 10^{-31})
Riemann primon gas	$z P(\beta)$, $P \sim \log(1/\varepsilon)$	temp-driven, log	$R \rightarrow 0^+$, $\sim 1/(L+\kappa)^2$, $\Delta_3 = 5.0451882$
all-integer Bose gas	$z(\zeta(\beta) - 1) \sim 1/\varepsilon$	temp-driven, pole	$R \rightarrow 0^+$, $R/\varepsilon^2 \rightarrow \Delta_3 = 5.693982$
ideal Bose gas (BEC)	$u^{-3/2} \text{Li}_{5/2}(e^y)$	fugacity-driven	$ R \rightarrow \infty$, $\sim \tau^{-1/2}$

Table 1: The dichotomy, bracketed. $u = \beta$; $\tau = -\log z \rightarrow 0^+$ at condensation; $\kappa = 0.832503$ for the primon gas. The two flat instances differ in spectrum (primes vs. all integers) and singularity type (log vs. pole); two of the four contain no primes.

All-integer Bose gas. $\prod_{n \geq 2} (1 - z n^{-\beta})^{-1}$, i.e. $\mathcal{P} = \zeta - 1$, along $z = 1$: $R/\varepsilon^2 = 4.95, 5.61, 5.69$ at $\varepsilon = 10^{-2}, 10^{-3}, 10^{-4}$, converging to $\Delta_3^{\text{int}} = 5.693982$.

5.1 The fugacity-driven complement: ideal Bose gas

To exhibit the divergent branch we evaluate the ideal Bose gas, where the singularity is fugacity-driven: as $\tau \rightarrow 0$ the macroscopic occupation of the ground state controls the fluctuations. Table 2 reports the computed scalar curvature on approach to BEC. The product $R \tau^{1/2}$ settles toward ≈ 0.11 , consistent with the analytically-derived leading divergence $|R| \sim \tau^{-1/2} \rightarrow \infty$ (the finite- τ effective exponent approaches $\frac{1}{2}$ from below through $O(\tau^{1/2})$ corrections).

Remark 3 (a geometric invariant of the primon gas). The renormalized information length of the Riemann primon gas from the Hagedorn point to $\beta = 2$,

$$C = \int_1^2 \left[\sqrt{\partial_\beta^2 \log \zeta(\beta)} - \frac{1}{\beta-1} \right] d\beta = -0.0343561541791219860831108814584476 \dots,$$

$\tau = (T - T_c)/T_c$	R	$R \tau^{1/2}$
0.3	0.3275	0.1794
0.1	0.4681	0.1480
0.03	0.7511	0.1301
0.01	1.2135	0.1214
0.003	2.1196	0.1161
0.001	3.5874	0.1134

Table 2: Scalar curvature of the ideal Bose gas approaching BEC ($z = e^{-\tau}$, sphere-positive convention): $|R| \rightarrow \infty$, leading $\sim \tau^{-1/2}$.

is, to 90 digits, not an integer combination of $\{\pi, \log 2, \log 3, \gamma, \gamma_1, \zeta(3), \zeta(5), G, \log A\}$ at coefficient height $\leq 10^6$ (PSLQ); we register it as a new constant of the gas.

6 Discussion

The dichotomy demonstrates that the thermodynamic geometry of arithmetic gases discriminates strictly by the source of the singularity. Because the singular sector of a temperature-driven transition factorizes as $e^y \varphi(x)$, it generates an exactly degenerate (flat) contribution to the Hessian determinant; the surviving curvature is sourced entirely by the finite analytic background coupling to that singular mode, and $R \rightarrow 0$.

Conversely, fugacity-driven transitions avoid this cofactor cancellation, permitting the standard Ruppeiner divergence $|R| \rightarrow \infty$. The general rate amplitude $R \sim [\Delta_3(\alpha + 1)/(2A^2)]\varepsilon^{2\alpha}$ (Appendix A) shows the flatness holds across arbitrary pole orders $\alpha > 0$ and the marginal logarithmic limit. We stress the precise status: the flatness is *unconditional* within the class — it survives even the degenerate background $\Delta_3 = 0$, with R vanishing one power faster ($\varepsilon^{2\alpha+1}$); the positive spectral condition $\Delta_3 > 0$, automatic whenever $\mathcal{P}(k) = \sum_q q^{-k} > 0$, fixes only the explicit rate *amplitude*. Thus the Hagedorn transitions of arithmetic gases are not merely physically distinct from ordinary condensation, but geometrically flat — and this is a statement of statistical mechanics: two of the four bracketing instances (the classical ideal gas and the all-integer Bose gas) involve no primes, and nothing here bears on the location of any zeta zero.

A Curvature rates: the cofactor reduction

Evaluate on $z = 1$, $\beta = 1 + \varepsilon$, $\varepsilon \rightarrow 0^+$, and abbreviate the singular (single-particle) derivatives $p_j := \mathcal{P}^{(j)}(1 + \varepsilon)$, $j = 0, 1, 2, 3$.

A.1 Singular/background split and the row reduction

By Lemma 2 the $k = 1$ term is the only singular one; write each partial as its $k = 1$ (singular) part plus the finite $k \geq 2$ background. From (2) at $z = 1$ the singular parts are $\psi_{xx}^s = p_2$, $\psi_{xy}^s = -p_1$, $\psi_{yy}^s = p_0$, $\psi_{xxx}^s = -p_3$, $\psi_{xxy}^s = p_2$, $\psi_{xyy}^s = -p_1$, $\psi_{yyy}^s = p_0$, so the three rows of N have singular parts

$$\text{Row}_1^s = (p_2, -p_1, p_0), \quad \text{Row}_2^s = (-p_3, p_2, -p_1), \quad \text{Row}_3^s = (p_2, -p_1, p_0) = \text{Row}_1^s.$$

The equality $\text{Row}_3^s = \text{Row}_1^s$ is Lemma 1 in coordinates: the $k = 1$ weight $k^{a+b-1} = 1$ is independent of b . Subtracting Row 1 from Row 3 annihilates the divergent singular part,

leaving the finite background differences

$$\text{Row}_3 - \text{Row}_1 = (\Delta_{xx}, \Delta_{xy}, \Delta_3), \quad \Delta_{xx} = \sum_{k \geq 2} k(k-1)\mathcal{P}''(k), \quad \Delta_{xy} = -\sum_{k \geq 2} k(k-1)\mathcal{P}'(k), \quad \Delta_3 = \sum_{k \geq 2} k(k-1)\mathcal{P}(k). \quad (4)$$

Determinant-invariance under $\text{Row}_3 \rightarrow \text{Row}_3 - \text{Row}_1$ gives

$$N = \Delta_{xx}M_{31} - \Delta_{xy}M_{32} + \Delta_3M_{33}, \quad (5)$$

with minors on rows 1, 2 whose leading (singular $\times$ singular) parts are

$$M_{31} \sim p_1^2 - p_0p_2, \quad M_{32} \sim p_0p_3 - p_1p_2, \quad M_{33} \sim p_2^2 - p_1p_3, \quad (6)$$

and $\det g \sim p_0p_2 - p_1^2$.

A.2 Pole singularity of order α

For $\mathcal{P}(1 + \varepsilon) \sim A\varepsilon^{-\alpha}$, $\alpha > 0$, $A \neq 0$: $p_0 \sim A\varepsilon^{-\alpha}$, $p_1 \sim -A\alpha\varepsilon^{-\alpha-1}$, $p_2 \sim A\alpha(\alpha+1)\varepsilon^{-\alpha-2}$, $p_3 \sim -A\alpha(\alpha+1)(\alpha+2)\varepsilon^{-\alpha-3}$. Substituting,

$$\det g \sim A^2\alpha\varepsilon^{-2\alpha-2}, \quad M_{33} \sim -A^2\alpha^2(\alpha+1)\varepsilon^{-2\alpha-4},$$

and since the minor orders are $M_{31} \sim \varepsilon^{-2\alpha-2}$, $M_{32} \sim \varepsilon^{-2\alpha-3}$, $M_{33} \sim \varepsilon^{-2\alpha-4}$, the M_{33} term dominates (5): $N \sim \Delta_3M_{33} = -\Delta_3A^2\alpha^2(\alpha+1)\varepsilon^{-2\alpha-4}$. Hence

$$R = -\frac{N}{2(\det g)^2} \sim \frac{\Delta_3A^2\alpha^2(\alpha+1)\varepsilon^{-2\alpha-4}}{2A^4\alpha^2\varepsilon^{-4\alpha-4}} = \frac{\Delta_3(\alpha+1)}{2A^2}\varepsilon^{2\alpha} \xrightarrow{\varepsilon \rightarrow 0} 0. \quad (7)$$

The singular powers cancel between N and $(\det g)^2$; the surviving $\varepsilon^{2\alpha}$ forces flatness.

A.3 Logarithmic singularity (marginal $\alpha \rightarrow 0$)

For $\mathcal{P}(1 + \varepsilon) = \log(1/\varepsilon) + a_0 + \dots$, $L = \log(1/\varepsilon)$: $p_0 \sim L$, $p_1 \sim -\varepsilon^{-1}$, $p_2 \sim \varepsilon^{-2}$, $p_3 \sim -2\varepsilon^{-3}$, and the cancellations of A.2 are replaced by logarithmically-enhanced leading terms, $\det g \sim (L + \kappa)/\varepsilon^2$, $M_{33} \sim -\varepsilon^{-4}$, $N \sim -\Delta_3/\varepsilon^4$, where κ collects the $O(1)$ singular-subleading and singular $\times$ background contributions ($\kappa = 0.832503$ for the primon gas). Hence

$$R \sim \frac{\Delta_3}{2(L + \kappa)^2} \xrightarrow{\varepsilon \rightarrow 0} 0, \quad (8)$$

the $\alpha \rightarrow 0^+$ boundary of (7): $\varepsilon^{2\alpha} \rightarrow 1$ ceases to vanish and the logarithmic growth of $\det g$ supplies the vanishing instead.

A.4 Conditions and the rate claim

Two hypotheses were used. (C1) (definite leading order) holds for any Beurling/prime spectral zeta. (C2) $\Delta_3 \neq 0$ is needed only for the explicit amplitude; since $\Delta_3 = \sum_{k \geq 2} k(k-1)\mathcal{P}(k)$ with $\mathcal{P}(k) = \sum_q q^{-k} > 0$, $\Delta_3 > 0$ automatically for every physical gas.

Remark 4 (flatness is unconditional). If $\Delta_3 = 0$ (non-physical), the M_{33} term drops and N is led by $\Delta_{xy}M_{32} \sim \varepsilon^{-2\alpha-3}$, giving $R \sim \varepsilon^{2\alpha+1} \rightarrow 0$ — one power faster. Thus $R \rightarrow 0$ for *every* temperature-driven singularity in the class, independent of (C2); only the amplitude depends on Δ_3 .

A.5 Numerical confirmation

Eq. (7) predicts $R/\varepsilon^{2\alpha} \rightarrow \Delta_3(\alpha + 1)/(2A^2)$. On the tunable-order family $\mathcal{P}(s) = (s - 1)^{-\alpha} 2^{1-s}$ ($A = 1$):

α	$\varepsilon = 10^{-2}$	10^{-3}	10^{-4}	predicted
1	2.708	2.968	2.997	3.000
2	2.402	2.526	2.538	2.540
3	2.366	2.531	2.549	2.551

The marginal case (8) is confirmed by the primon gas ($R(L + \kappa)^2 \rightarrow \Delta_3/2 = 2.5226$) and the $\alpha = 1$ case by the all-integer Bose gas ($R/\varepsilon^2 \rightarrow \Delta_3 = 5.6940$).

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